ELL305 – Computer Architecture

ELL305 Term Paper – Draft 1

Work Done as of 24 August 2026

Provisional Topic

Modern Processor Design Beyond Monolithic Scaling: Chiplet Architectures and 3D-Stacked Cache in AMD Zen Processors

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This submission is an independent milestone progress report. It records the state of the work as of 24 August 2026 and is not intended to be a preliminary version of the final 150-page term paper.

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1 Chosen Topic and Present Scope

The proposed term paper studies how modern processor design is moving beyond the traditional assumption that most major processor resources must be integrated on a single large monolithic die. The principal case study will be AMD's Zen family of processors, with particular emphasis on two developments: chiplet-based processor organization and three-dimensional stacked last-level cache through AMD 3D V-Cache technology.

The present working research question is:

> **How do chiplet-based organization and 3D-stacked cache help modern processors address limitations in performance, scalability, memory latency, power, and die-level integration, and what new trade-offs do these techniques introduce?**

The intention is not to write a product-history comparison of AMD and Intel. Instead, AMD Zen will be used as a modern case study through which processor-organization and memory-system concepts from ELL305 can be investigated in greater depth. A limited comparison with contemporary Intel approaches may later be included where it helps clarify alternative architectural choices.

The topic has also been selected with the 150-page final-paper requirement in mind. A report restricted to one particular processor or benchmark would probably be too narrow for such a length. The broader formulation permits the final paper to develop the theoretical background – processor organization, performance, cache hierarchy, multicore systems, interconnects, power, and scaling – before studying their application in modern chiplet and 3D-cache designs.

2 Mapping to the ELL305 Textbook

The primary textbook is Smruti R. Sarangi's *Basic Computer Architecture*, Version 3.09 [1].

An initial mapping of the proposed work to the course text has been completed. The strongest connections are to Chapters 9–12.

2.1 Chapter 9 – Processor Design

The chapter provides the basic model of processor organization, including the fetch, execution and memory-access units, data path, and control path. This will serve as the baseline processor model before modern partitioned and chiplet-based implementations are considered.

2.2 Chapter 10 – Principles of Pipelining

The relevant topics include pipelining, performance metrics, advanced processor techniques, and power/temperature considerations. These concepts provide a framework for studying processor performance and for understanding why increasing clock frequency alone cannot indefinitely provide performance scaling.

2.3 Chapter 11 – The Memory System

This is presently the strongest direct mapping to the cache component of the proposed paper. The chapter introduces temporal and spatial locality, cache hierarchy, cache design, cache misses, hit time, miss penalty, and memory-system optimization. These concepts provide the theoretical basis for studying the motivation and performance implications of increased last-level-cache capacity and 3D V-Cache.

2.4 Chapter 12 – Multiprocessor Systems

The chapter introduces private and shared caches, cache coherence, banked caches, multicore organization, and interconnection networks. These ideas are useful when extending the discussion from a conventional multicore processor toward systems containing multiple compute dies connected through an interconnect fabric.

As advised, I intend to consult Prof. Sumantra Dutta Roy, since the proposed work maps primarily to material contained within Chapters 9–12. No formal consultation has taken place yet. The consultation will be used to confirm the scope and determine how far the term paper should enter topics such as advanced packaging, cache coherence, and die-to-die interconnection.

3 Preliminary Technical Survey

A preliminary literature and technical survey has been initiated to determine whether the topic has sufficient technical depth and reliable primary material for the final term paper.

One important source identified is Samuel Naffziger, Kevin Lepak, Milam Paraschou, and Mahesh Subramony's paper, "AMD Chiplet Architecture for High-Performance Server and Desktop Products," presented at ISSCC 2020 [2]. This source is particularly relevant because it discusses the architectural and engineering motivation behind AMD's chiplet approach, including the increasing difficulty and cost associated with large leading-edge dies and the limited benefit of advanced process scaling for many I/O structures.

A second source identified is the AMD Zen 2 presentation by Suggs et al. at Hot Chips 31 [3]. Zen 2 is important to the proposed study because the compute-die/I/O-die organization became central to both mainstream Ryzen and second-generation EPYC processors. The material provides an initial basis for studying the Core Complex Die (CCD), I/O Die (IOD), and AMD Infinity Fabric.

For the cache component, a key primary source identified is John Wuu et al.'s ISSCC 2022 paper, "3D V-Cache: The Implementation of a Hybrid-Bonded 64MB Stacked Cache for a 7nm x86-64 CPU" [4]. This work is directly relevant because it links the familiar computer-architecture problem of increasing effective cache capacity with a modern physical-integration method in which additional cache SRAM is vertically stacked over the processor rather than being implemented entirely in the planar CPU die.

AMD's technical documentation for the Zen 3 EPYC generation has also been shortlisted [5]. It gives a useful architectural description of the relationship between Zen 3 cores, the Core Complex, the Core Complex Die, and the shared L3 cache.

At this stage, these references have been examined at a **preliminary level**: bibliographic details, abstracts, architectural summaries, and selected technical material. A structured source-by-source reading log will be maintained during the next phase.

4 Preliminary Research Directions

The initial survey suggests that the chiplet discussion should not be reduced to the observation that a processor contains several pieces of silicon. The important architectural questions concern **what is partitioned, why the partition is made, and what costs are introduced by moving communication across die boundaries**.

The chiplet portion of the paper is therefore expected to examine subjects such as:

- Core Complex Dies and I/O Dies;
- Infinity Fabric and die-to-die communication;
- latency and bandwidth;
- process-node specialization;
- manufacturing and die-size considerations;
- scalability with core count;
- interconnect power; and
- cache organization and coherence.

The 3D V-Cache portion has an especially direct connection to Chapter 11. A hierarchical memory system attempts to satisfy a large fraction of memory accesses in relatively small, fast caches. Increasing last-level-cache capacity can reduce expensive accesses to lower levels for workloads that benefit from the additional working-set capacity. However, conventional cache expansion also involves area, latency, power, and implementation costs.

3D V-Cache is therefore interesting as an architectural case study because it modifies part of this trade-off: additional SRAM capacity can be integrated vertically through a stacked cache die.

This leads to several questions for later study:

- Under what workload conditions does additional L3 capacity provide substantial performance improvement?

- How should increased cache capacity be evaluated against access latency and clock frequency?
- Why are some workloads more cache-sensitive than others?
- What thermal and power constraints arise when cache is vertically stacked?
- How does the cache hierarchy interact with the chiplet organization of the processor?
- To what extent are observed performance gains attributable to cache, core microarchitecture, frequency, memory bandwidth, or other factors?

5 Work Packages and Progress

WP	Task	Progress
WP1	Topic selection, scope refinement, research question, and mapping to Chapters 9–12	75%
WP1	Initial textbook orientation and selective review of relevant processor, performance, cache, and multiprocessor sections	20%
WP2	Literature-search strategy and identification of technical/primary sources	30%
WP2	Bibliographic verification and collection of accessible primary source material	5%
WP2	Structured reading of shortlisted references	10%
WP3	Focused AMD case study: Zen 2 to Zen 3 to X3D development	5%
WP4	Careful comparative study of Intel and/or alternative approaches	0%
WP5	Experimental or simulation methodology, if appropriate	0%
WP6	Long-form synthesis, subsequent drafts, revision, and final 150-page paper	0%

The percentages are approximate because this first milestone occurs very early in a semester-long project. The present progress is concentrated primarily on defining a viable research problem, mapping it to ELL305, and identifying an appropriate primary literature base.

6 Provisional Semester Schedule

The semester plan presently consists of the following phases:

- **Late August–mid September:** complete the textbook foundation and organize the research material;
- **Late August–September:** expand and systematically read the literature;
- **September:** conduct a focused study of Zen 2 chiplet organization and Infinity Fabric;
- **Late September–October:** study Zen 3 cache organization and 3D V-Cache;
- **October–early November:** conduct comparative and quantitative analysis;
- **October–November:** complete long-form section writing and synthesis; and
- **November:** integrate the sections, audit citations, revise, and prepare the final submission.

Since official dates for Drafts 2 and 3 have not yet been announced, this schedule is provisional and will be updated after the course milestones are published.

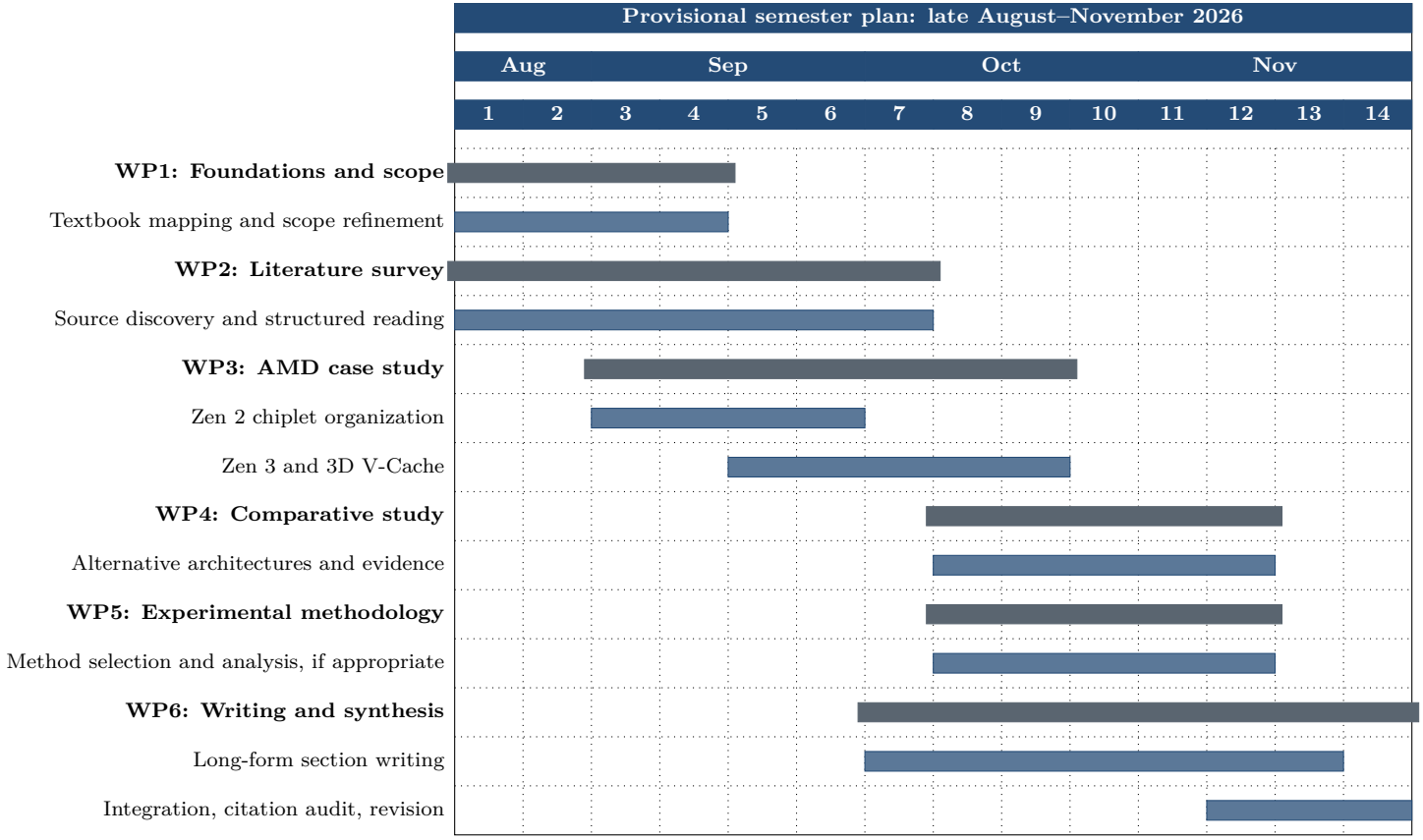


Figure 1: Provisional work-package schedule. Week numbers are relative planning units and will be revised when later milestone dates are announced.

7 References Identified and Examined So Far

The initial source set consists of the primary course textbook, a Hot Chips presentation on Zen 2, two peer-reviewed ISSCC papers on AMD's chiplet and stacked-cache designs, and AMD's technical overview of EPYC 7003 processors. These sources are listed because they have been identified and examined at a preliminary level; inclusion here does not imply that a complete close reading has already been performed.

Initial bibliography

[1] S. R. Sarangi, *Basic Computer Architecture*, Version 3.09. Author's online edition, 2025.

[2] S. Naffziger, K. Lepak, M. Paraschou, and M. Subramony, "AMD chiplet architecture for high-performance server and desktop products," in *2020 IEEE International Solid-State Circuits Conference (ISSCC)*, IEEE, 2020, pp. 44–45. DOI: 10.1109/ISSCC19947.2020.9063103

[3] D. Suggs, D. Bouvier, M. Clark, K. Lepak, and M. Subramony, "AMD Zen 2," in *2019 IEEE Hot Chips 31 Symposium (HCS)*, IEEE, 2019.

[4] J. Wuu et al., "3D v-cache: The implementation of a hybrid-bonded 64MB stacked cache for a 7nm x86-64 CPU," in *2022 IEEE International Solid-State Circuits Conference (ISSCC)*, IEEE, 2022, pp. 428–429.

[5] Advanced Micro Devices, Inc., "AMD EPYC 7003 processor microarchitecture overview," Advanced Micro Devices, Inc., 57075, 2022. [Online]. Available: https://www.amd.com/content/dam/amd/en/documents/epyc-technical-docs/white-papers/overview-amd-epyc7003-series-processors-microarchitecture.pdf

8 Immediate Next Steps

The immediate next tasks are to consult Prof. Sumantra Dutta Roy; complete the first focused textbook pass, beginning with Chapter 11 and selected parts of Chapters 10 and 12; import and organize the initial references; maintain one structured note per major source; and expand the literature search to independent research on chiplet systems, die-to-die interconnects, 3D integration, cache scaling, and memory-system performance.

After this first literature-review stage, a decision will be made on whether the final paper should contain an original small-scale cache simulation or benchmark-based analysis in addition to the literature-based AMD case study.